



**Department of Management**  
**Research Methods in Business**  
**Chapter I:**  
**An Introduction**

## **I.1. Definition and Basic concepts**

**Research**: (re-search =“search again”) is defined as the systematic and objective process of generating information from data (gathering, organizing, analyzing and interpreting data) to enhance decision making

# Characteristics of Research

1. **Systematic:** it is neither intuitive nor haphazardly gathered
2. **Objective:** impersonal rather than biased
3. **Useable (purposive):** It should be done to facilitates the managerial decision process.
4. **Rigor**-based on sufficient theoretical ground and sound methodology.
5. **Replicability**- can be conducted again and should also testable.
6. **Precision**-closeness of the finding to reality and the tolerable error should be indicated.
7. **Generalizability**- a research finding should be applicable to different organizations with modifications.

## **Based on purpose of its finding**

### **A. Basic/pure / fundamental research**

- to **expand the limits** of knowledge.
- **Not directly** involved in the solution to a practical problem.
- Findings generally cannot be immediately implemented.
- Deals with **theoretical aspect** of science-Investigate **new theory** or **verify existing**

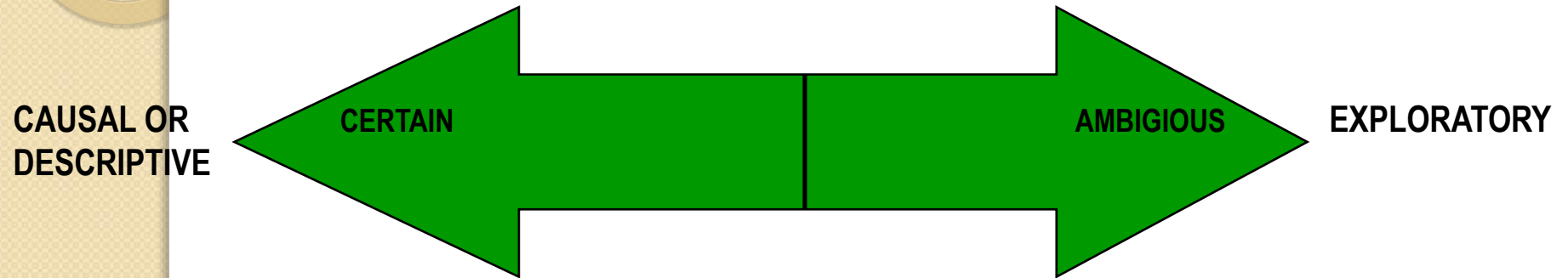
Ex: Do managerial success is correlated with level of education?

## B. Applied Research

- To solve an existing problem
- Conducted when a decision must be made about a specific real-life problem.
- Discover ways to apply theories to solve existing problems.
- EX: to evaluate the effectiveness of particular production method.

2. Based on Degree of problem definition/ purpose

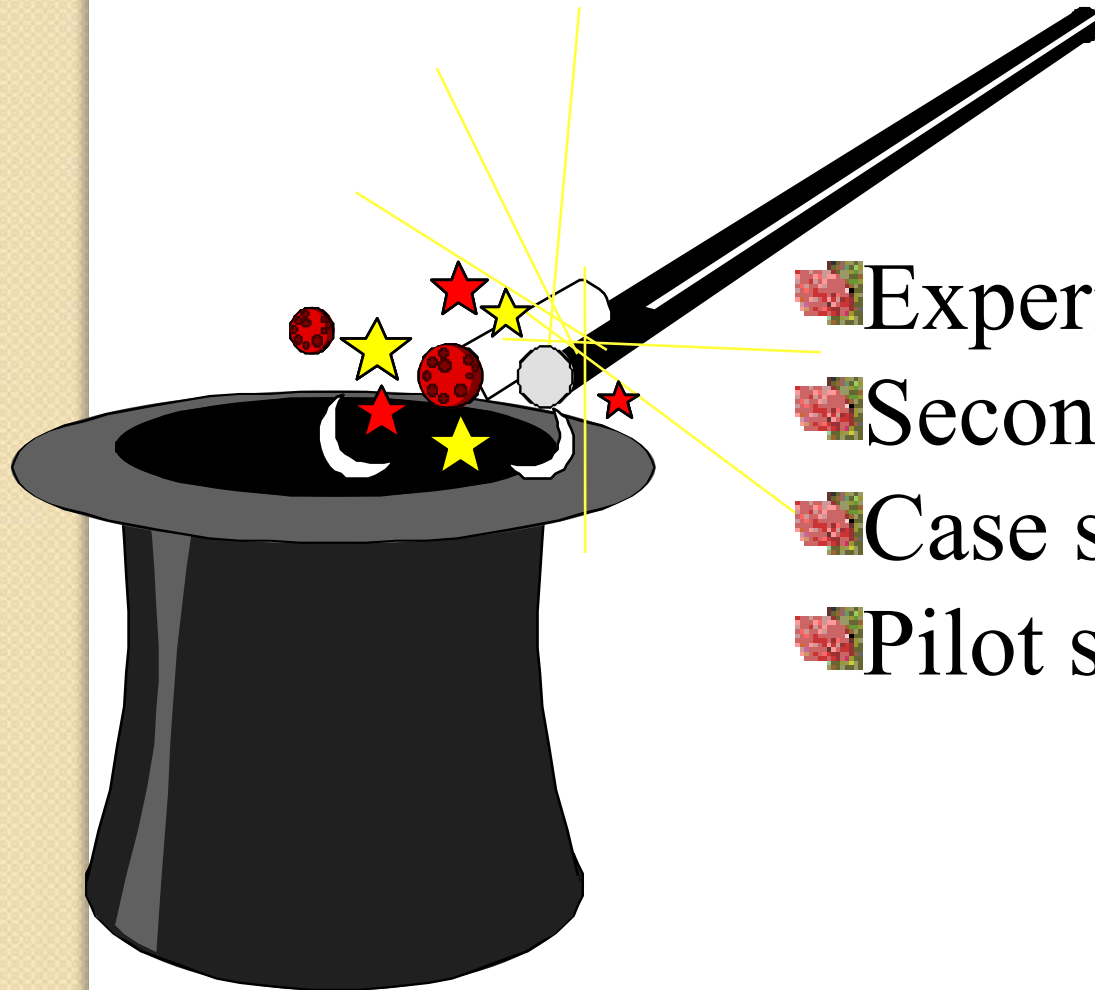
➤ uncertainty influences the type of research



# **I. Exploratory /preliminary research**

- Conducted to clarify **ambiguous problems**.
- is conducted with the expectation that subsequent research will be done.
- Done to learn something about the study area(esp when it is new).
- Its purpose is to investigate the general situation in a particular field

# Techniques to explore



- Experience surveys
- Secondary data analysis
- Case studies
- Pilot studies

# Experience Surveys

- Ask knowledgeable individuals about a particular research problem
  - most are quite willing



“If you wish to know the road up the mountain, you must ask the man who goes back and forth on it.”

Zenrinkusi





# Secondary Data Analysis

- Data collected for a purpose other than the project at hand
- Economical
- Quick source for background information

# Case Study Method

- Intensely investigates one or a few situations similar to the problem
- Investigate in depth
- Careful study
- May require cooperation
- The results should be seen as tentative

# Pilot Study

Any small scale exploratory study that uses sampling, But does not apply rigorous research methods.

EX: Focus Group Interviews

In-Depth Interviews



## 2. Descriptive research:

- Done to describe characteristic of a population or phenomenon as it exist.
- Seek to determine the answers to **who, what, when, where and how questions.**
- A researcher has no control over variables.
- can be case study or survey design

## Descriptive...

- The primary interest is to understand the conditions or relationships that exist, opinions that are held, processes that are going on, effects that are evident or trends that are developing.
- It is possible to make comparisons or contrasts and attempt to discover relationships between existing non-manipulated variables.
- Correlational research is one form of descriptive research and its goal is to determine whether two or more variables are related.
- For example, a researcher may be interested in determining whether age is related to weight.

# Descriptive cont'd...

- There are two phases in a descriptive study:
  - a) Descriptive** : where the researcher merely describes the variables or events relevant to a given problem.
  - b) Analytical** : this phase goes a step beyond mere description. It includes the study of association of one variable with another. It attempts to indicate what happens when two or more factors interact in a given situation.

### 3. Causal/ Explanatory research:

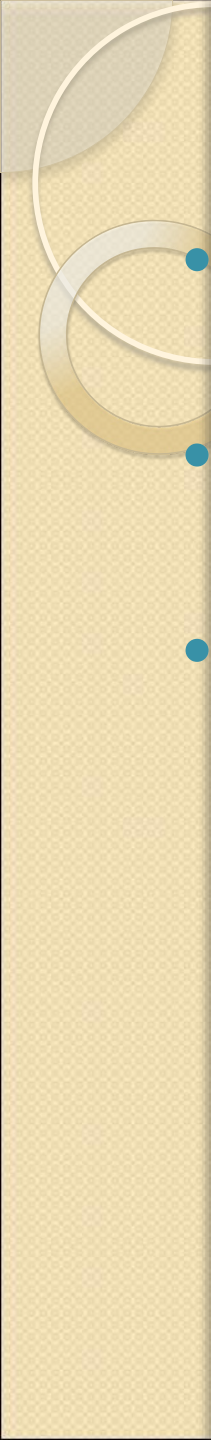
- the main goal is to identify **cause-and-effect** relationships among variables.
- Answers why variables are the way they are and used to determine the association between variables.
- e.g. Influence of advertising on profits,

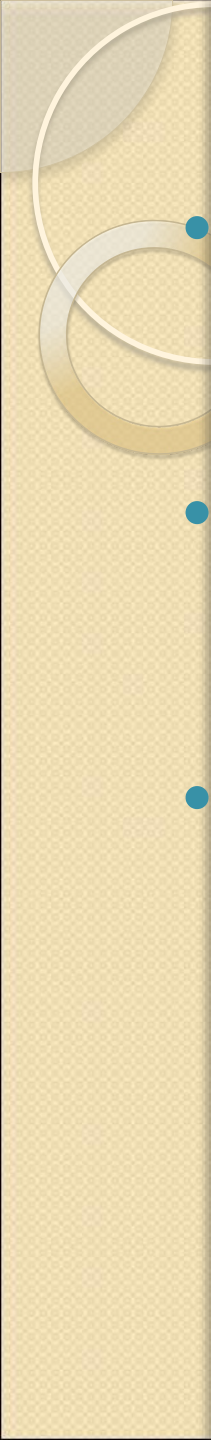
# Explanatory...

- Explanatory can be of the **experimental design** or of the **non-experimental design**.
- Research in which the **independent variable is manipulated** is termed 'experimental research' and
- A research in which an **independent variable is not manipulated** is called 'non-experimental research or **ex-post facto**.'

# Experimental research

- This is the type of research in which **variables are manipulated** and their effects upon other variables are observed.
- Experimental research provides a systematic and logical method for answering the question, “If this is done under carefully planned and controlled conditions, what will happen?”
- In other words, experimental research shows what will be the outcome when certain variables are carefully controlled.

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- There are three prerequisites for drawing an inference of causality between two variables:
  - First, there must be a relationship (i.e., a correlation) between the two events.
  - In other words, the events must *co-vary as one changes, the other must also change*. If two events do not co-vary, then a researcher cannot conclude that one event caused the other event.

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- Second, one event (the cause) must precede the other event (the effect)-sometimes referred to as a *time-order relationship*. This should make intuitive sense.
  - Obviously, if two events occur simultaneously, it cannot be concluded that one event caused the other.
  - Similarly, if the observed effect comes before the presumed cause, it would make little sense to conclude that the cause *caused the effect*.

- 
- Third, alternative explanations for the observed relationship must be ruled out.
  - Stated another way, a causal explanation between two events can be accepted only when other possible causes of the observed relationship have been ruled out.

### **C. On the basis of data used**

- 1. Qualitative research:-** applicable for phenomenon that cannot be expressed in terms of labels and expressions
  - mainly use inductive approach
- 2. Quantitative research:-** based on the measurement of quantity or amount. It is applicable for phenomenon that can be expressed in terms of quantity(numerical & statistical data).
  - mostly use deductive approach



**D. On the basis of time required to complete the research**

**1. Cross sectional/One time research:-**  
research limited to a single time period.

✓ Example: customer satisfaction of new product

**2. Longitudinal/time series research: -**  
research carried out over several time periods.

✓ Use time series data

✓ Example: impact assessment

## 1.3. The scientific method

- The scientific method is a set of research principles and methods that helps researchers obtain valid results from their research studies.
- Because the scientific method deals with the general *approach* to research rather than the *content* of specific research studies, it is used by researchers in all different scientific disciplines

# Characteristics of the scientific method

- Although some disagreement exists regarding the exact characteristics of the scientific method, most agree that it is characterized by the following elements:
  - Empirical approach
  - Observations
  - Questions
  - Hypotheses
  - Experiments
  - Analyses
  - Conclusions
  - Replication

# Empirical Approach

- The *empirical approach* is an evidence-based approach that relies on direct observation and experimentation in the acquisition of new knowledge.
- Data derived from direct observation and experimentation.

# Observations

- In this sense, *observation* refers to two distinct **concepts**:
  1. Being aware of the world around us and
  2. Making careful measurements.

# Questions

- After getting a research idea, perhaps from making observations of the world around us, the next step in the research process involves translating that research idea into an answerable question.

# Hypotheses

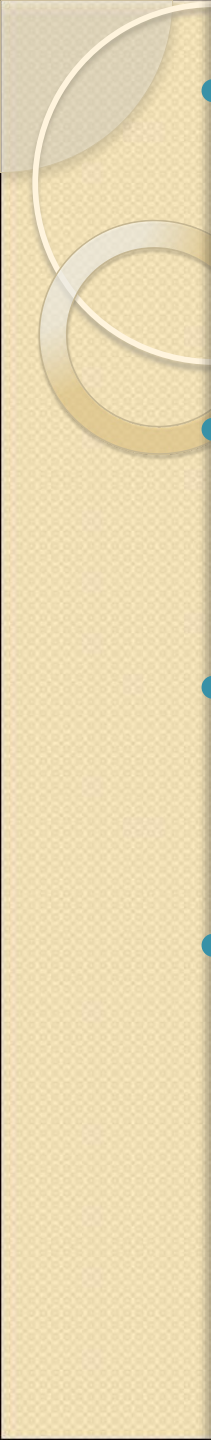
- A hypothesis is simply an educated and testable guess about the answer to your research question.
- Hypotheses are derived from the observations and relationships accepted as facts
- A hypothesis is a statement of what you think is true based on available evidence. Your research will then set out to prove or disprove the validity of this hypothesis.

# Experiments

- After articulating the hypothesis, the next step involves actually conducting the experiment (or research study).
- The researcher would design and conduct a study that would attempt to address that question.
- A key aspect of conducting a research study is measuring the phenomenon of interest in an *accurate* and *reliable* manner

# Analysis

- After conducting the study and gathering the data, the next step involves analyzing the data, which generally calls for the use of statistical techniques.
- The type of statistical techniques used by a researcher depends on the design of the study, the type of data being gathered, and the questions being asked.
- Although a detailed discussion of statistics is beyond the scope of this course, it is important to be aware of the role of statistics and software in conducting a research study.

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- Statistics help researchers minimize the likelihood of reaching an erroneous conclusion about the relationship between the variables being studied.
  - A key decision that researchers must make with the assistance of statistics is whether the null hypothesis should be rejected.
  - Remember that the null hypothesis always predicts that there will be no difference between the groups.
  - Therefore, rejecting the null hypothesis means that there *is* a difference between the groups.

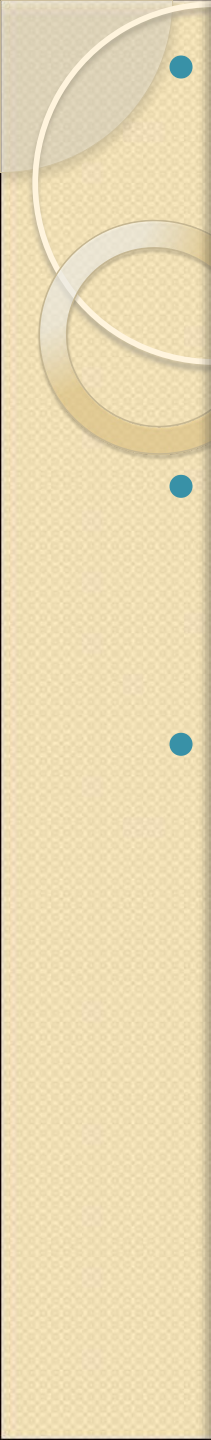
# Conclusions

- After analyzing the data and determining whether to reject the null hypothesis, the researcher is now in a position to draw some conclusions about the results of the study.
- For example, if the researcher rejected the null hypothesis, the researcher can conclude that the phenomenon being studied had an effect a *statistically significant* effect, to be more precise.

# Replication

- One of the most important elements of the scientific method is replication.
- *Replication* essentially means conducting the same research study a second time with another group of participants to see whether the same results are obtained

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- Replication illustrates an important point about scientific research researchers should **avoid drawing broad conclusions based on the results of a single research study** because it is always possible that the results of that particular study were an aberration.
  - In other words, it is possible that the results of the research study were obtained by chance or error and, therefore, that the results may not accurately represent the actual state of things.
  - However, if the results of a research study are obtained a second time (i.e., replicated), the likelihood that the original study's findings were obtained by chance or error is greatly reduced.

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- Replication serves several integral purposes, including establishing the reliability (i.e., consistency) of the research study's findings and determining whether the same results can be obtained with a different group of participants.
  - This last point refers to whether the results of the original study are *generalizable* to other groups of research participants.
  - If the results of a study are replicated, the researchers—and the field in which the researchers work—can have greater confidence in the reliability and generalizability of the original findings.



## **1.4 Research process:**

Research follows a general pattern what is described by its process. The stages are:

1. Defining the research problem
2. reviewing extensive literatures
3. Determining basic methodologies
  - deciding research design and type
  - determining data type and source
  - determining methodology used



3. Deciding sample

4. Collecting data- instruments, time needed, nature of data required, who collect and other aspects should be clarified here.

5. Analyzing the data

6. Report writing



## **I.5.The motives of doing research**

**one or more of the following:**

1. Desire to get a **research degree**
2. Desire to face the **challenge in solving the unsolved problems**
3. Desire to get **intellectual joy** of doing some creative work
4. Desire to be of **service to society;**
5. Desire to get **respectability** Etc

## Chapter 2 Research problem & hypothesis

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- What is Research Problem?
- Approaches to Problem Definition
- Research Questions
- Formulating Specific Research Questions
- Variables and Hypotheses Formations
- Research Objectives

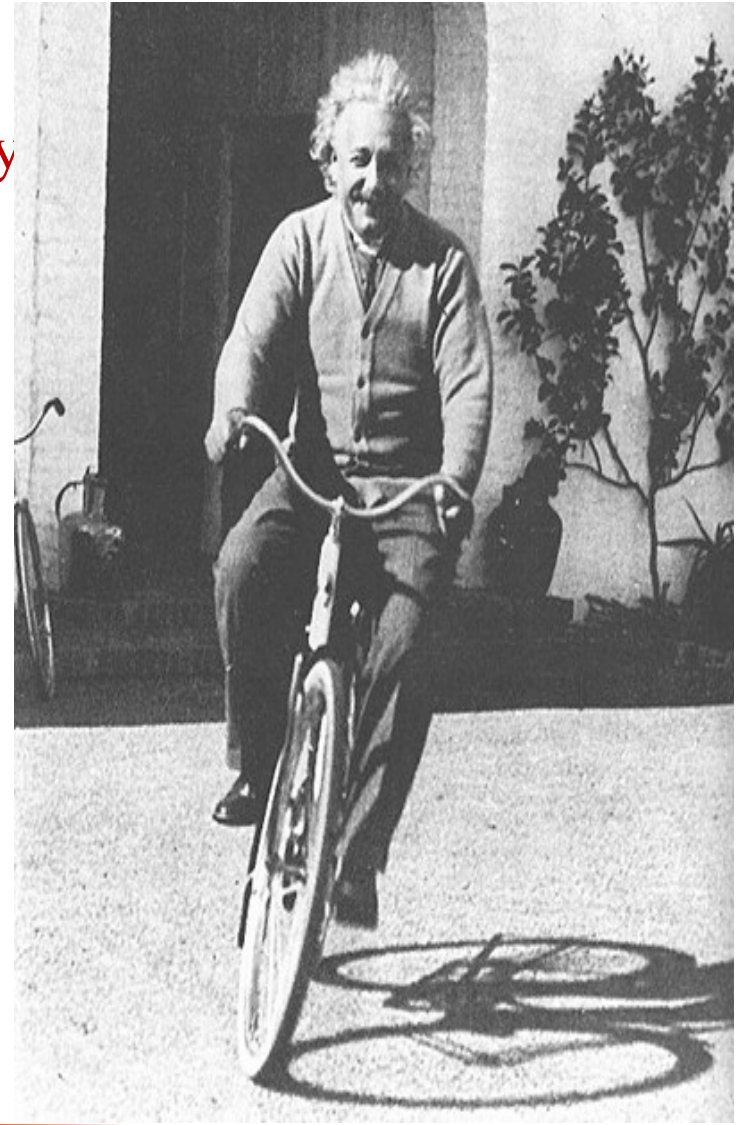
## 2.1 Research problem definition

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- ♠ **It** refers to some **difficulty** which a researcher experiences in the **theoretical** or **practical situation** and wants to obtain **a solution for it**.
- ♠ **A problem definition** *indicates a specific decision area to be clarified or problem to be solved*
- ♠ *It specifies the **research questions** to be answered and **the objective of the research to be achieved**.*
- ♠ Understanding the true nature of the problem is a foundation for definition of research problem.

## 2.1 Definition research problem (CONT...)

1. "A problem well defined is half solved" **John Dewey**
2. "If I had only one hour to save the world, I'd spend 55 min. defining the problem and 5 minutes finding a solution."
3. "The formulation of the problem is often more essential than its solution." (Albert Einstein)



## 2.1 Definition research problem (Contn...)

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- ♠ Thus, defining a research problem properly is a prerequisite to conduct a study.
  - ♠ A well defined research problem enable to determine:
    - **What data** are to be collected? Nature of data
    - **where to be collected?** Sources of data
    - **How to collected data?** Instruments
    - **How to analyze the data?** Statistics, Software packages, models etc.
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## 2.2. The Process of Problem Definition

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### **The Process of Problem Definition includes:**

Step 1 : Understand the Background of the Problem

Step 2: Isolate and Identify the Problems, Not the Symptoms

Step 3: Determine the Relevant Variable

*Step 4: State research questions/Hypothesis & research objectives*

Step 5: Determine the Unit of Analysis

## Step I : Understand the Background of the Problem

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It includes a **Situation analysis** :

- **Preliminary informal** gathering of background information to familiarize researchers with the decision area
- Doing **environmental scanning/observation**
- Discussing with those **who have a good knowledge** of the problem concerned
- Conducting pilot study etc.

## Step 2: Isolate and Identify the Problems, Not the Symptoms

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### Example :

Brewery company- consumer prefer taste of competitor's product

- **Problem definition:**

Taste of brewery product needs to be reformulated

- **True problem :**

old fashioned package is influencing taste perception.

## Step 3: Determine the Relevant Variable

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### A Variables:

- Is anything that **varies or changes in value**
- **Assume different numerical values** usually in magnitude or strength, or in direction.
- It can be:
  - Continuous/ quantitative
  - Categorical/qualitative
  - Dependent
  - Independent

## Types of variable

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### I. Quantitative variables:

- involves measurable numerical or statistical data
- Can be **Discrete or Continuous**
- **Discrete variables** : expressed in integer values, they are non-continuous variables and usually it is a result of counting.  
E.g Age, family size e.t.c
- **Continuous variables**: assumes any range of values and usually result from measurement.  
Eg. Sales volume, profits etc.

### 2. Qualitative/ Categorical Variables:

- Is one that indicates membership in some group
- attitude, behavior, feeling, and satisfaction etc
- It can be quantified:
  - presence or absence of the concerning attribute(s) or
  - by level of the attributes.

eg1. available equals 1 and not available 0

eg2. Level of satisfaction

### 3. Dependant variable :

- Is the primary interest to the researcher.
- The researcher's goal is to understand and describe the dependant variable, or to explain its variability or to predict it.

GRQ: **What is the effect of service level on sales volume ?**

- **Dependant variable = sales volume**

Service level



Sales

### 4. Independent Variable:

- is one that **influences the dependant variable** in either a **positive or negative** way.
- That is, when the independent variable changes, the dependent variable is also changes (with each unit of change in the independent variable, there is a change in the dependent variable also).
- also known as the **predictor variable**
- Example: service level

Service level



Sales

## Dependant and independent variable relationships

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- Positive relationship:
  - Two variables move in the same directionExample:
- Negative relationship:
  - Two variables move in opposite direction
  - Example

### 5. Moderating Variable:

- The moderating variable is the one that has a strong **contingent effect on the independent-dependent relationship**. That is, the presence of a third variable (the moderating variable) modifies the original relationship between the independent variable and the dependent variable.

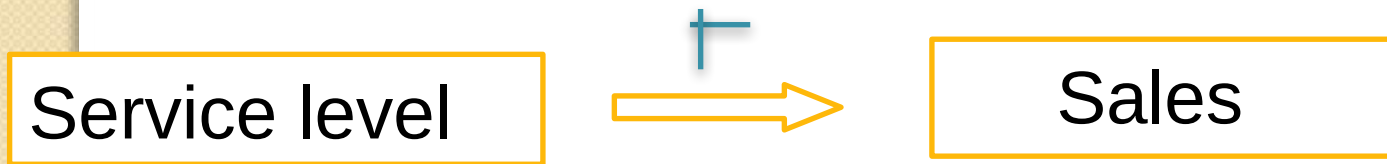
## 6. Intervaning variable

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**For example:**

**dependent vs. independent variable**

What is the effect of service level on sales?



For example:

Independent Vs intervening Vs Dependant variable

GRQ: What is the effect of service level on sales?

- SRQ1: What is the effect of service level on **customer satisfaction**?
- SRQ2: What is the **effect of customer satisfaction** on sales?



## **Research questions**

- assume two forms:
  1. A central question / general research question
    - Consists of several specific research questions
  2. Associated sub questions (specific research questions)
    - Are segments/parts of the GRQ
    - Answers to all SRQ's make up answer to GRQ

## Specific research questions: Example

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- GRQ: Which factors influence the **quality** of tomato's during transport?
  - SRQ1: How does the **temperature** during transport influence the quality of the tomatoes?
  - SRQ2: How do the **atmospheric** conditions during transport influence the quality of the tomatoes?
  - SRQ3: How do **mechanical conditions** during transport influence the quality of the tomatoes?

## Specific research questions: Example

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### ■ For example:

GRQ: What is the effect of service level on sales?

- SRQ1: What is the effect of service level on customer satisfaction?
- SRQ2: What is the effect of customer satisfaction on sales?

## **Step 5: State the research questions ( Hypothesis) and research objective**

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### **Research Hypothesis**

- *Hypothesis is an unproven proposition or possible solution to a problem which can or may not be real solutions*
- *Declarative statements that can be empirically tested.*
- They provide the guidelines for the type of data and techniques necessary for analysis.
- Hypotheses are important bridges between empirical inquiry and theory

### **Forms of hypothesis**

#### **a. If---then**

- If employees are healthier, then they will take sick leave less frequently.
- If worker's attitudes towards an organization climate are changed in positive direction, there will be an increase in organizational effectiveness

### ***b. Directional and non-directional Hypotheses***

If, in stating the r/ship b/n variables or comparing two groups, **terms such as positive, negative, more than, less than** and the like are used, then these hypotheses are directional because the direction of the r/ship b/n the variables is indicated.

### **Example**

- There is a r/ship b/n age and job satisfaction

### C. Null and alternative hypothesis

#### Null hypothesis

- It states that the population correlation b/n two variables is equal to zero or that the difference in the mean of two groups in the population is equal to zero (or some definite number).
- It expresses as no (significant) r/ship b/n two variables or no (significant) difference b/n two groups.

#### Alternate Hypothesis

- It is, the opposite of null, a statement expressing a r/ship b/n two variables or indicating d/ces b/n two groups.
  - By contrast, the *alternate hypothesis* predicts that there will be a difference between the groups.
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# Characteristics of a Good Hypothesis

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- ♠ ***Clear And Precise.***
  - ♠ ***Capable Of Being Tested***
  - ♠ State ***Relationship Between Variables***
  - ♠ ***Be Limited In Scope And Must Be Specific.***
  - ♠ Stated As Far As Possible In ***Most Simple Terms***
  - ♠ Consistent ***With Most Known Facts,***
  - ♠ Hypothesis Should Be Open To ***Testing Within A Reasonable Time.***
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# Research objectives

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- The research objective describes your motivation to do the research
  - what do you hope to achieve with your research?
- Often stated in two forms
  - broad (general) and
  - specific research objectives.

## **step 5: Determine the unit of analysis**

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- ❖ Unit of analysis indicates what or who should provide the data and at what level of aggregation.
- ❖ The investigation may focus on the collection of data about organizations, departments, work groups, individuals or objects
- ❖ Individuals, households, organizations, etc.

# Chapter 3

## Research Design

- ❖ *Research design is the plan and structure of the investigation.*
- ❖ *The research design constitutes the blueprint for the collection, measurement and analysis of data.*
- ❖ Research design focuses upon turning a research question and objectives into a research project.
- ❖ It considers :
  1. Research type choice
  2. Research design/strategy choice

## Choice of research methods type

- ✦ The power of the researcher to affect the variables under study (experimental and the ex-post facto).
- ✦ The purpose of the study & degree to which a research problem is crystallized (descriptive or causal study).
- ✦ The time dimension (cross-sectional or time series study).
- ✦ The research environment (field, laboratory or simulation).
- ✦ Nature of data used:(Quantitative or qualitative )
- ✦ Based on method (s) employed: *Mono Method* using a single data collection & corresponding analysis method or *multiple methods* using more than one data collection & analysis methods)

## Research Design Choice

1. Experimental design
2. Survey/census design
3. Case Study design

### **Note:**

- *Each strategy can be used for different types of research.*
- *No research strategy is inherently superior or inferior to any other*
- *Your choice of research strategy will be guided by*
  - ✓ *your research question(s) and objectives,*
  - ✓ *the extent of existing knowledge*
  - ✓ *the amount of time and other resources you have available*
- *Research strategies are not mutually exclusive, rather can be used together*

# I. Experimental Design

- The purpose of an experiment is to study causal links; whether a change in independent variable produces a change in another variable
- *Experimental research differ from other research strategies in terms of degree of control over variables.*
- Experiments tend to be used in exploratory and explanatory research to answer 'how' and 'why' questions.
- It can be **basic experimental designs** (a **single independent** variable is manipulated to observe its effect on a **single dependent** variable) or **factorial experimental designs** allow for investigation of **two or more independent variables on dependent variable.**

- 1. Manipulation of the independent variable (s)*
- 2. Experimental treatments and control groups*
- 3. Selection and measurement of the dependant variable*
- 4. Selection and assignment of test units*
- 5. Control over extraneous variables*

# **1. Manipulation of the independent variable (s)**

- ✓ **Independent variables:-** is the variable that can be manipulated & its value may be changed or altered independently of any other variable.

# **2. Experimental treatments and control groups**

- ✓ **Experimental treatments:-** manipulation of the independent variable being investigated
- ✓ **Experimental group:-** a group of subjects exposed to an experimental treatment
- ✓ **Control Group:-** a group of subjects who are exposed to the control condition (not exposed to the experimental treatment)

### **3. Selection and measurement of the dependent variable**

- ✓ *Its value depends on experimenter's manipulation of the independent variable.*
- ✓ *The criterion or standard by which the results are judged.*

### **4. Selection and assignment of test units**

- ✓ *Subjects or entities whose responses to the experimental treatment are measured or observed.*
- ✓ *Individuals, organizational units, sales territories or other entities may be the test units.*

### **5. Control over extraneous variables**

- *Variables that may affect the dependent variables other than identified dependent variable(s).*
- *Such variables must be controlled.*

# ***Ways of Controlling Extraneous Variables***

1. ***Randomization***: you want to make the result of the experiment wholly governed by your **treatment and the laws of chance**.
2. ***Increase the size of the experiment*** either through the provision of more replicates or by the inclusion of additional treatments.
3. ***Refine the experimental technique*** by securing uniformity in the selection of experimental materials and the application of the treatments.
  - ✓ Control external influences so that every treatment produces its effect under comparable and desired conditions

## 2. Survey Research strategy

- Survey is a research technique in which data is gathered from a sample of respondents using verbal or written questioning
- It is most frequently used to answer who, what, where, how much and how many questions.
- Usually used for exploratory and descriptive research.
- Respondents are a representative of population(sample).
- Gathering Information via Surveys is quick, inexpensive, efficient & provides flexibility.

# Survey can be cross-sectional or longitudinal

Longitudinal survey can be subdivided into:

1. **Trend studies** A given general population is studied at different points in time while the specific population & samples included are different in each survey.
  - ✓ Same general population-d/t specific population & sample
2. **Cohort studies** Collects data from the same specific population each time data are collected although the sample respondents studied are different.
  - ✓ Uses the same general, specific population to measure changes at over time but samples included are d/t.
3. **Panel studies** Involves collecting data from the same sample of individual respondents over time.
  - ✓ Same general, specific population but sample is d/t

- A. Random Sampling Error-** A statistical fluctuation that occurs because of change in variables of elements selected for the sample
- B. Systematic Error (bias)-** Systematic error results from some imperfect aspect of the research design or from a mistake in the execution of the data.

Includes:

1. Respondent error
2. Administrative error

# **1. Respondent error**

- a. Non response- not available, no interest etc
- b. Response bias: Deliberate falsification & Unconscious misrepresentation
  - ✓ Acquiescence bias
  - ✓ Extremity bias
  - ✓ Interviewer bias
  - ✓ Auspices bias
  - ✓ Social desirability bias

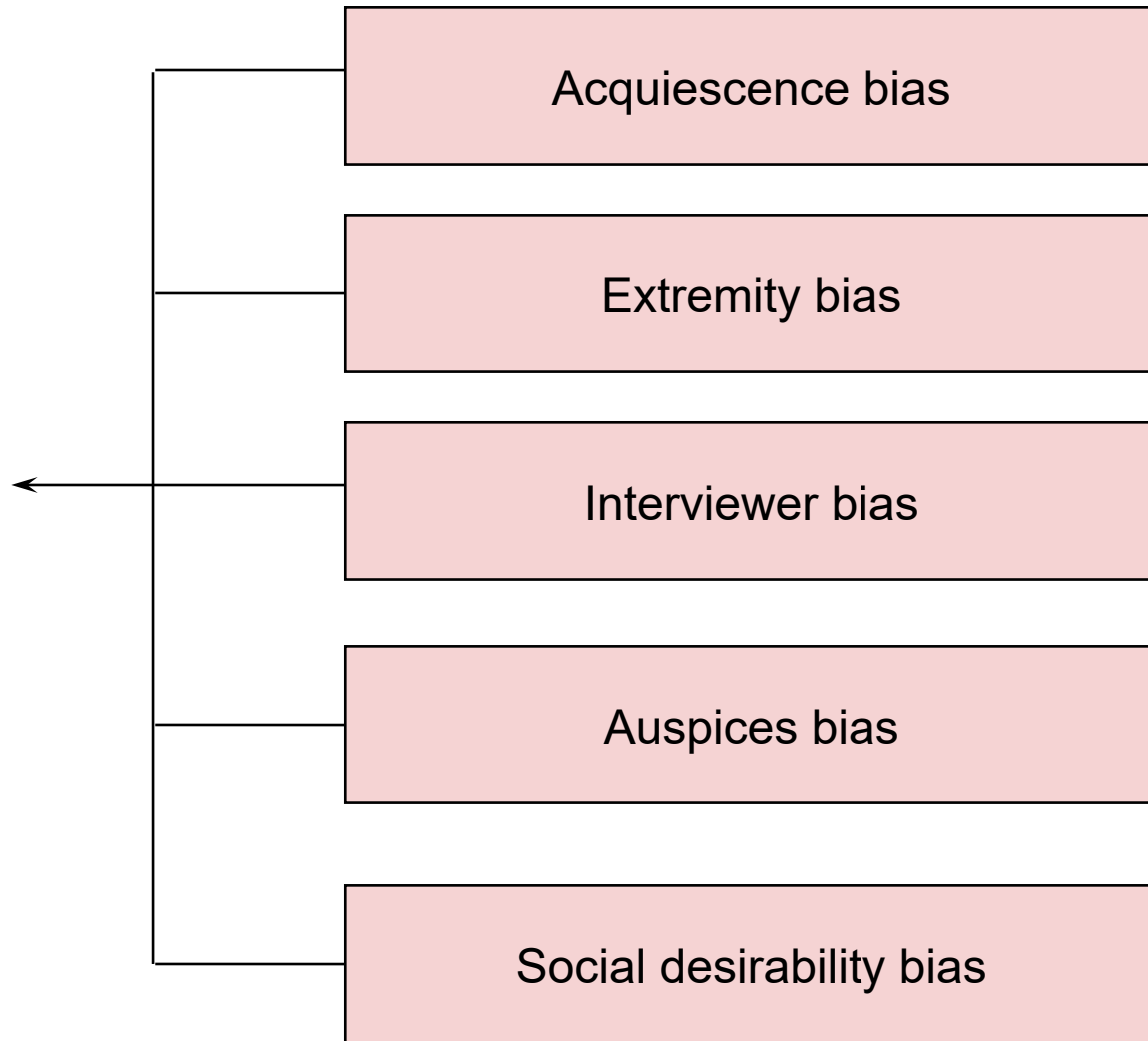
## **2. Administrative error**

- Data processing error
- Sample selection error
- Interviewer error
- Interviewer cheating

# I. Respondent Error

- A classification of sample bias resulting from some respondent action or inaction
  - a. **No response error:** no respondents - people who refuse to cooperate or not-at-home.
  - b. **Response bias:** a bias that occurs when respondents tend to answer questions with a certain slant that **consciously or unconsciously misrepresents the truth.**

## Cont'd



### i) Acquiescence Bias (yes, yes, yes)

- A category of response bias that results because some individuals tend to agree with all questions or to concur with a particular position.

### ii) Extremity Bias

- A category of response bias that results because response styles vary from person to person; some individuals tend to use extremes when responding to questions.

### iii) Interviewer Bias

- A response bias that occurs because the presence of the interviewer influences answers.

## Cont'd

### iv) Auspices Bias

- Bias in the responses of subjects caused by the respondents being influenced by the organization conducting the study.

### v) Social Desirability Bias

- Bias in responses caused by respondents' desire, either conscious or unconscious, to gain prestige or appear in a different social role.

## II. Administrative Error

- Improper administration of the research task such as mistakes (Confusion, Neglect & Omission)
- Includes:
  - **Data processing error** - incorrect data entry, computer programming, or other procedural errors during the analysis stage.
  - **Sample selection error** - improper sample design or sampling procedure execution.
  - **Interviewer error** - field mistakes
  - **Interviewer cheating** - filling in fake answers or falsifying interviews

### 3. Case study

- A strategy for doing research which involves an empirical investigation of a particular contemporary phenomenon within its real life context using multiple sources of evidence (Robson,2002)
- Often used to answer the question ‘why?’ as well as the ‘what?’ and ‘how?’ questions.
- case study strategy is most often used in explanatory and exploratory research
- likely to need to use and triangulate multiple sources of data

**Triangulation** refers to the use of different data collection techniques within one study

Case study strategies:

- **A single case** is often used where it represents a critical case considered for the study.
- **Multiple case:** more than one case.
  - The rationale for using multiple cases focuses upon the need to establish whether the findings of the first case occur in other cases and, as a consequence the need to generalise from the finding.

## Cont'd

- ❖ **Holistic case** research is concerned only with the organisation as a whole
- ❖ **Embedded case** study: concerned with sub-units within the organisation, perhaps departments or work groups.

# Chapter 4: Research Proposal

- ❖ It is a written statement of the research plan that includes a statement explaining the purpose of the study and a detailed, systematic outline of a particular research methodology to be followed.
- ❖ It presents the problem to be researched & its importance
- ❖ To discuss the research efforts of others who have worked on related problems.
- ❖ Shows data type necessary for solving the problem
- ❖ Shows how data will be gathered & analyzed

# Research proposal cont'd...

- Is the road map of the researcher which reveals about
  - ❖ What will be done
  - ❖ Why it will be done
  - ❖ How it will be done
  - ❖ Where it will be done
  - ❖ To whom it will be done, and
  - ❖ What is the benefit of doing it?

# Basic functions of research proposal

- It serves as:

## **A. a means of communication**

- Between the investigator and other stakeholders ( consultants, sponsors, etc)

## **B. a plan**

- Provides an inventory of what must be done and what materials have to be collected as preliminary step.

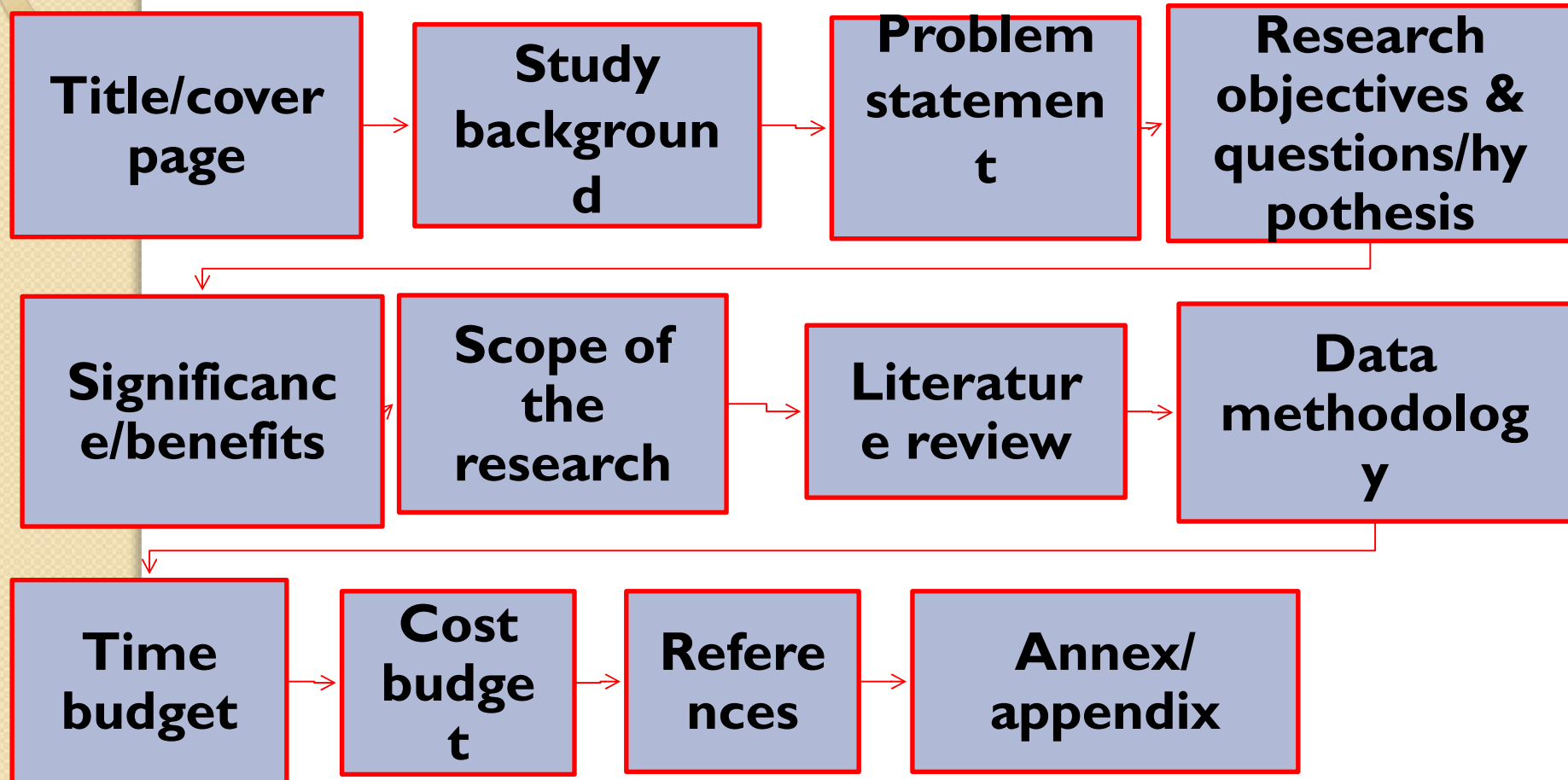
## **C. a contract**

- A bond of agreement between parties

# **Formats and elements of research proposal**

- There is no hard and fast rule in writing a specific scientific research proposal
- It varies depending on the company and the complexity of the research.
- Though formats vary, the essences expressed in the different formats remain the same

# Elements of Research Proposal



# I. Title (or Cover) Page

- reflects the research under consideration
- must be *relevant, feasible, applicable* and *cost-effectiveness* .
- It should be clear, short, and transparent
- It should capture and reflect the content of the proposal.
- It should enable the readers to understand the concepts of the study.

## **2. Background of the study**

- It encompasses background information/what is known so far about the study area.
- It contains relevant information about the study area either in inductive or in deductive way.
- It should be concise and informative.

# 3. Problem statement

- Consists reason for conducting the study.
- It should discuss the management dilemma and the question, its significance, and why something should be done to change the status quo.
- It is the foundation for the development of a proposal (i.e., of objectives, methods, etc...)
- It allows clarification of the topic selected
- Be sure your problem statement is clear without the use of idioms or clichés.

## 4. Research objectives

- addresses the purpose of the investigation.
- deals with exactly what is being planned by the proposed research.
- In a descriptive study, the objectives can be stated as the research question.
- If the proposal is for a descriptive or causal study, then the objectives can be restated as a hypothesis.
- an objective for the research proposal must be **SMART (Specific, Measurable, Achievable, Realistic, and time-bound)**.

## **5. Research Questions**

- It is a question form of specific objectives
- It should correspond with specific objectives
- It should be specific and operational

## **6. Hypothesis**

- It is used as a tentative solution and it will be tested via research process.
- Like specific objectives and research questions it should be specific.
- It is widely used as substitute with research question

**It**

## **7. Significances/Benefits of the Research**

- Benefits to be achieved by undertaking the research will be discussed in terms of:
  - ❖ Benefits to the researcher
  - ❖ its benefits to management/organization,
  - ❖ Its benefits to the sponsor,
  - ❖ Its benefits to society
  - ❖ its importance to serve as an input for a further detailed study will

## 8. Literature Review

- A research literature review is a **systematic**, **explicit**, and **reproducible** method for **identifying**, **evaluating** and **synthesizing** the existing body of completed and recorded work produced by researchers, scholars, and practitioners.

## Tips in literature review

- Locating literature from a variety of sources
- Reading it carefully and thoroughly, and
- Paraphrase it thoroughly
- Organizing it into themes (ideas) along the line of investigation.

## Importance of literature review

- To identify the potential difficulty in the investigation
- To avoid duplication ( maintain originality)
- To improve the methodological and procedural matters, and
- Identify the gap in the specific study topic.

## **Literatures are reviewed to answer:**

- ✓ What is already known about this area?
- ✓ What concepts and theories are relevant?
- ✓ What research methods and strategies have been used in this area?
- ✓ Are there any controversies?
- ✓ Are there any inconsistencies in findings?
- ✓ Are there unanswered questions?

## 9. Methodologies of the study

- the core and the largest part of a scientific research proposal
- is an essential prerequisite for validating the results and hence acceptability of the proposal
- should give full details to show how the research activity is going to be carried out:
- **Method of sampling, data collection & analysis**

## Data

- information required to attack the problem is explained
- Each variable or indicator identified by a researcher is explained and justified to show how it is relevant to the current study
- The source of the data, site (location) of the study and time that will be included (relevant time) should also be specified

## **I 0. Time plan**

- is a schedule, chart, or graph that summarizes in a clear fashion various components of a research undertaking as per its time

## **I I. Cost plan**

- It may be convenient to use the work plan as a starting point
- Specify for each activity in the work plan what resources are required
- Determine for each resource the unit cost and the total cost
- Keep some amount for contingency

## 12. References

- References are listed alphabetically by the author's last name.
- References should be selected based on their relevance

## In the text reference

- IN the case of Ethiopian names, the author's given (first) name precedes that of the father's name, e.g. Solomon Kassa and not Kassa S.- Ethiopian names should not be abbreviated.

## In the text reference cont'd...

- (Abebe Kebede and Ketema Hailu 1989)
- (Hartmann and Kester, 1975, andersson et al. 1993, Darwin and Morgan, 1993) chronologically.

According to Abebe Kebede and Ketema Hailu (1989),...

- For three or more authors, use et al. (not italics) i.e. Abiy Astatke et al. ( 1989) in the text ( but spell out all authors names in the reference list).

# Formats for listing references in the reference section

- **Journal article**

- Mahli, S.S Harapiak, J.Nybiorg M. and N.A Flore. 1991. Soil chemical properties after long term N fertilization of broom grass nitrogen rate. Communications in soil science and plant analysis 22 1447-1458.
- Gezahegn Ayele and Tekalign Mamo 1995. Determinants of demand for fertilizer in vertisol cropping system in Ethiopia. Tropical Agriculture (Trinidad) 72: 165-169

# Book

- Chapman, D.H. and P.F. pratt. 1961. Methods of Analysis for soils and waters. University of California, Riverside, California.(N.B initials appear before last authors family name).

## Chapter in book

- Loegering, W.Q. 1984 \. Genetics of the pathogen host association pp. 165-192. In William R. Bushnell and Alan P. Roelfs (eds) The cereal rusts, Vol 1. Academic press Orlando Florida.

## Paper in proceedings

- Mesfin Abebe. 1992. An investigation into the cause of wilt in cotton. pp. 129-139. In : proceedings of symposium on cotton production under irrigation in Ethiopia Melka Werer, Ethiopia 21-22 October 1982, Institute of Agricultural Research Addis Ababa

# Unpublished materials

- Citation of unpublished and other source materials not readily available in libraries should not be included in the reference list but should be mentioned in parentheses in the text or as a footnote.

# I 3. Appendix/Annex

- It is not mandatory to have this section
- Information types provided in this section are those additional details, which are difficult to accommodate within the standard headings.
- Staffs to be included in appendix are:
  - Details questionnaires and interview schedule
  - Survey and other geographic maps, etc...

# I. Problem

- The problem is not of sufficient importance or is unlikely to produce any new or useful information.
- The proposed research is based on a hypothesis that rests on insufficient evidence, is doubtful, or is unsound.
- The problem is more complex than the investigator appears to realize.
- The problem is scientifically premature and warrants, at most, only a pilot study.
- The description of the nature of the research and of its significance leaves the proposal nebulous and diffuse and without a clear research aim.

# Cont'd.....

## 2. Approach

- proposed tests, or methods, or scientific procedures are unsuited to the stated objective.
- The description of the approach is too nebulous, diffuse, and lacking in clarity to permit adequate evaluation.
- The overall design of the study has not been carefully thought out.
- The statistical aspects of the approach have not been given sufficient consideration.
- The material the investigator proposes to use is unsuited to the objective of the study or is difficult to obtain.
- The number of observations is unsuitable.

# Cont'd.....

## **3. Investigator**

- The investigator does not have adequate experience or training for this research.
- The investigator appears to be unfamiliar with recent pertinent literature or methods.
- The investigator's previously published work in this field does not inspire confidence.
- The investigator proposes to rely too heavily on insufficiently experienced associates.

# Chapter 5

## Sample Design and Sampling Procedures

### 5.1 Terms and Definitions

- Population: Any complete group of People, object etc under study.
- Elements/cases: Members of the population.
- Census: Investigation of all individual elements that make up a population
- **Sample**: Subset/portion of a larger population.
- **Sampling**: the process of gathering data from some portion of the population.
  - Sampling enables to make **Statistical inference**

## 5.2 Needs for sampling

1. Budget and time Constraints (in case of large populations)
2. Physical impossibility of checking all elements
3. Destructive nature of some census
4. Adequacy of sample results: (if sample is representative of population no need of census)

# Sampling Techniques

The sampling techniques available to you can be divided into two types:

- Probability or representative sampling;
- Non-probability or subjective sampling
- For many research projects, you will need to use a variety of sampling techniques at different stages
- Some projects involving both probability and non-probability sampling techniques.

## 5.3. Probability Sampling

- the probability, of each element being selected from the population is known & is usually equal for all.
- probability sampling is often associated with survey and experimental research strategies.

**The process of probability sampling can be divided into four:**

1. Identify a suitable sampling frame based on your research objectives.
2. Decide on a suitable sample size.
3. Select the most appropriate sampling technique and select the sample.
4. Check that the sample is representative of the population

### 5.3.1. Identification of Sampling Frame

- The **sampling frame** for any probability sample is a complete list of all the cases/elements in the population from which your sample will be drawn.

Examples:

- the list of companies on the stock exchange,
- the directory of medical doctors and specialists,

## Cont'd

### Selecting your sampling frame : Checklist

- ✓ Are cases listed in the sampling frame relevant to your research topic, in other words will they enable you to answer your research question and meet your objectives?
- ✓ How recently was the sampling frame compiled, in particular is it up to date?
- ✓ Does the sampling frame include all cases, in other words is it complete?
- ✓ Does the sampling frame contain the correct information (is it accurate)?
- ✓ Does the sampling frame exclude irrelevant cases (is it precise)?

## 5.3.2 Deciding on a suitable sample size

Choice of sample size is governed by:

1. the confidence you need to have in your data and thereby the findings
2. the margin of error that you can tolerate that is, the accuracy you require for any estimates made from your sample;
3. Variability observed in the target population, and to lesser extent
4. the types of analyses you are going to undertake
5. the size of the total population from which your sample is being drawn

## Remarks

- sample size is determined by:
  - *relevant formula* to the study,
  - based on previous *similar study*,
  - *rule of thumb/percentage of the total population*
  - *consider all (census)* if population size is limited.
- In all of the above ways of sample size determination, adjustments are possible.
- a minimum number of 30 for statistical analysis (central limit theorem)

## ***Sample size for continuous data***

$$n = \frac{(z_{\alpha/2})^2 \sigma^2}{e^2}$$

Where:  $\delta$  = *standard deviation*

$e$  = *margin of error*

$z_{\alpha/2}$  = *the z value of alpha divided by 2*

$n$  = *sample size*

# Sample size for categorical data/proportion

$$n = \frac{(z_{\alpha/2})^2 p(1-p)}{e^2}$$

*Where*

*$n$  = sample size*

*$P$  = probability of success*

*$1-p$  = probability of failure*

## 5.3.3 Selecting appropriate sampling Technique

Five main techniques can be used to select a probability sample:

- ☐ Simple random;
- ☐ Systematic random ;
- ☐ Stratified random;
- ☐ Cluster random and
- ☐ Multi-stage random.

# ***1. Simple Random Sampling***

 **Simple random sampling** involves selecting the sample at random from the sampling frame using either random number tables or lottery system

To do so,

1. Number each of the cases in your sampling frame with a unique number.
2. Select cases using random numbers until your actual sample size is reached.

## Cont'd

### **When to Use? It is recommended when:**

- ✓ Statistical inferences are to be made
- ✓ The population is geographically concentrated
- ✓ The sampling frame does not have relevant strata
- ✓ The sampling frame contains periodic patterns

## ***Advantages of Simple Random Sampling***

- ✓ only minimal advance knowledge of population needed
- ✓ Easy to analyze data and less processing error
- ✓ Simple to use and understand
- ✓ Requires only one stage of sample selection

# *Disadvantages Simple Random Sampling*

- ✓ Requires sampling frame to work from
- ✓ Does not use knowledge of population that researcher may have
- ✓ Larger errors for sample size than with stratified sampling
- ✓ Respondents may be widely dispersed , hence costs may be higher

## *2. Systematic sampling*

Systematic sampling involves you selecting the sample at regular intervals from the sampling frame. To do this you:

1. Number each of the cases in your sampling frame with a unique number.
2. Select the first case using a random number.
3. Calculate the sampling fraction.
4. Select subsequent cases systematically using the sampling fraction to determine the frequency of selection.

# When to use systematic sampling?

- ✓ Statistical inferences are to be made
- ✓ The population is geographically concentrated
- ✓ The sampling frame does not contain relevant strata
- ✓ The sampling frame does not contain periodic patterns

# Advantages and Disadvantages of systematic sampling

## Advantages

- ✓ Simple to draw samples
- ✓ Easy to check

## Disadvantages

- ✓ periodicity: a problem that occurs when the original list has a systematic pattern
- ✓ May introduced increased biased

### ***3. Stratified random sampling***

It involves dividing the population into two or more relevant and significant strata based on one or a number of attributes. To do so, it needs to:

1. Choose the stratification variable or variables.
2. Divide the sampling frame into the discrete strata
3. Number each of the cases within each stratum with a unique number, as discussed earlier
4. Select your sample using either simple random or systematic sampling, as discussed earlier.

## When to use stratified random sampling?

- ✓ Statistical inferences are to be made
- ✓ Research require does/does not require face to face contact
- ✓ The population is geographically concentrated
- ✓ The sampling contains relevant strata

# **Advantages and Disadvantages of stratified random sampling**

- Assures representation of all groups in sample
- Characterization of each stratum can be estimated and comparisons made
- Reduces variability for same sample size

## **Disadvantages**

- requires accurate information on proportion in each stratum
- If stratified lists are not already available, they can be costly to prepare

## 4. *Cluster sampling*

Cluster sampling is similar to stratified sampling as you need to divide the population into discrete groups prior to sampling

- The groups are termed **clusters** in this form of sampling and can be *based on any naturally occurring grouping*
- For cluster sampling your **sampling frame** is the complete list of clusters rather than a complete list of individual cases within the population

*When to use Cluster sampling?*

- ✓ Statistical inferences are to be made
- ✓ Research require does/does not require face to face contact
- ✓ The population is in discrete geographic clusters
- ✓ The sampling frame contains relevant clusters

# Pros and cons of *Cluster sampling*

## Pros

- ✓ Yields lowest field cost if clusters are geographically defined
- ✓ Requires listing of all clusters but of individuals only within clusters
- ✓ Can estimate characteristics of clusters as well as of population

## Cons

- ✓ larger error for comparable size than other probability samples
- ✓ Researcher must be able to assign population members to unique clusters
- ✓ Duplication or omission of individual elements may occur

## ***5. Multi-stage sampling***

- Development of cluster sampling
- It is normally used
  - ✓ to overcome problems associated with a geographically dispersed population
  - ✓ when face-to-face contact is needed
  - ✓ where it is expensive and time consuming to construct a sampling frame for a large geographical area
- like cluster sampling, you can use it for any discrete groups, including those that are not geographically based
- The technique involves taking a series of cluster samples, each involving some form of random sampling

## 5.4. **Non-probability sampling**

- the probability of each case being selected from the total population is not known
- It is impossible to address objectives that require you to make statistical inferences about the characteristics of the population
- When it is difficult to specify a sampling frame
- No appropriate statistical techniques for measuring random sampling error
- Often used in qualitative researches

## Stages in Non-probability sampling

1. Identify a suitable sampling frame based on your research question(s) or objectives.
2. Decide on a suitable sample size.
3. Select the most appropriate sampling technique and select the sample.
4. Check that the sample is representative of the population

# I. Identification of Sampling Frame

- ◆ generally used when there is no suitable sampling frame
- ◆ In ability or any other plausible reason to prepare sampling frame

## 2. Deciding on a suitable sample size

- ❖ unlike probability sampling, there are no rules
- ❖ the logical relationship between your sample selection technique and the research purpose is important
- ❖ your sample size is dependent on your research question(s) and objectives. In particular:
  - ✓ what you need to find out,
  - ✓ what will have credibility and
  - ✓ what can be done within your available resources

### **3. Selecting the most appropriate sampling technique and the sample**

Five main techniques can be used to select a probability sample:

1. Quota sampling;
2. Purposive sampling;
3. Snowball sampling ;
4. Self selection sampling;
5. convenience sampling

# *1. Quota sampling*

- Based on the premise that your sample will represent the population as the variability in your sample for various quota variables is the same as that in the population.
- selection of cases within strata is entirely non-random.

Procedures:

1. Divide the population into specific groups.
2. Calculate a quota for each group based on relevant and available data.
3. Combine the data collected by interviewers to provide the full sample

# Cont'd

## When to use

- ✓ Statistical inferences are not required.
- ✓ The sample is required to be representative
- ✓ Relevant quota variables are available

## Advantages

- ✓ Less costly and can be done quickly
- ✓ Does not require sampling frame
- ✓ Introduces some stratification of population

## Cont'd

- Introduces bias as interviewers may diffuse
- Sampling error can not be estimated
- Projecting data beyond sample is inappropriate

## *2. Purposive sampling*

- Purposive or judgemental sampling enables you to use your judgement to select cases that will best enable you to answer your research question(s) and to meet your objectives.
- Useful when
  - ✓ statistical inferences are not to be made
  - ✓ the sample is not required to be representative
  - ✓ no difficulty of identifying individual cases
  - ✓ the sample to be selected are very small
  - ✓ informative cases are sought for

# *Advantages and Disadv of Purposive sampling*

## **Advantages**

- ◆ ample guaranteed to meet a specific objective
- ◆ Easy to understand and undertake
- ◆ No sampling frame is required

## **Disadvantages**

- Bias due to experts' beliefs may make sample unrepresentative
- Projecting data beyond sample is inappropriate

### *3. Snowball sampling*

- Snowball sampling is commonly used when it is difficult to identify members of the desired population, for example people who are working while claiming unemployment benefit
- Procedures:
1. Make contact with one or two cases in the population.
  2. Ask these cases to identify further cases.
  3. Ask these new cases to identify further new cases (and so on).
  4. Stop when either no new cases are given or the sample is as large as is manageable

# *Advantages and Disadv of Snowball sampling*

## **Advantages**

- Useful in locating members of rare populations
- No sampling frame is required

## **Disadvantages**

- ◆ The main problem is making initial contact. Once you have done this, these cases identify further members of the population, who then identify further members
- ◆ problems of bias are huge, as respondents are most likely to identify other potential respondents who are similar to themselves, resulting in a homogeneous sample

## 4. *Self-selection sampling*

- Self-selection sampling occurs when you allow each case, usually individuals, to identify their desire to take part in the research.
- Often appropriate when the purpose of the investigation is just exploratory.
- Therefore it needs to:
  1. Publicise your need for cases, either by advertising through appropriate media or by asking them to take part.
  2. Collect data from those who respond

## *5. Convenience sampling*

- ◆ Convenience sampling (or haphazard sampling) involves selecting haphazardly those cases that are easiest to obtain for your sample
- ◆ The sample selection process is continued until your required sample size has been reached.
- useful when
  - ✓ there are little variation in the population
  - ✓ no difficulty of identifying individual cases
  - ✓ large sample is required

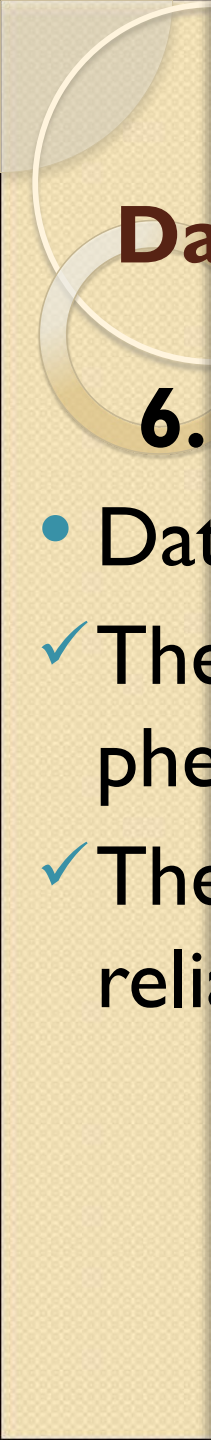
# *Advantages and Disadv of Snowball sampling*

## **Advantages**

- ◆ Easy to understand and undertake
- ◆ Very low cost
- ◆ No sampling frame is required

## **Disadvantages**

- prone to bias and influences that are beyond your control
- projecting beyond sample is inappropriate



# **Chapter 6**

## **Data Collection, Analysis & Interpretation**

### **6.1. Data and its source**

- Data are building blocks of research:
- ✓ The more data the researcher has about the phenomenon, the better will be the investigation.
- ✓ The more valid is the source of data, the more reliable will be

# Data sources

## ◆ Primary source

- Gathered by the researcher for the study on hand.
- The sources are individuals and the incidence around them, thus said to be original.
- Means: survey ,observation and experiment

## ◆ Secondary sources

- Includes all types of published and unpublished public or private documents and other such types of data.
- They have already gone through statistical process and are collected by someone else.
- Means: manuscripts, records, reports, letters, manuals etc.

# Why a researchers use data from secondary sources?

- **It has supplementary use:** the review of available secondary data provides a better understanding of the problem & it may also suggest solution.
- **It may solve problems:** if adequate data are available from secondary sources, primary data collection will not be necessarily required
- **Search costs are substantially lower:** it is cheap and less time consuming than primary sources.
- **It will help to plan the collection of primary data:** Examination of techniques employed by other investigators in similar studies may be useful in planning the present one.

## 6.2. Methods of Data Collection

### A. Interview

- Interview is a method of data gathering via open dialogue with respondent.
- Enables immediate feedback and clarification of doubts.
- The interview method can yield certain types of confidential information that an individual might be reluctant to put in writing.
- It can be personal or telephone interview; Can be structured or unstructured

- **Personal interview:** This is a face to face interaction in which the interviewer, asks a person being interviewed, the respondent, questions designed to obtain answers of the research problem.
- The strength of an interview lies in its flexibility, ensuring a high rate of response, control of the interview situation & recording of spontaneous responses.
- **Telephone interview:** Contacting respondents via telephone to gather responses to survey questions
- It is fast and lower cost, however it is difficult to observe the body language of respondent.

# ***Structured/unstructured interviews***

- ***Structured interviews:*** This is the use of standardized interview schedules that have been carefully prepared in advance to obtain information pertinent to the research problem.
  - The questions, their sequence and their wording are fixed.
- ***Unstructured Interviews:*** These are much more flexible. Although the research purposes govern the questions asked, their content, their sequence and their wording are entirely in the hands of the interviewer.

## B. Questionnaire

- It is a document containing questions and related items prepared to solicit information from respondents.
- The key factor in questionnaires design are:
  - ▶ Relevance of the questions to the objectives of the study & respondents.
  - ▶ Be simple, clearly stated and easy to answer
  - ▶ Contain only one major idea
  - ▶ It should not give clue or guide the respondent toward particular answer
  - ▶ Questions should be relevant to the central problem of the research

## C. Observation

- observation is the systematic process of recording the behavioral patterns of people, objects and occurrences as they are witnessed.
- No questioning of or communicating with people occurs.
- Observation becomes a tool for scientific inquiry when:
  - Systematically planned and executed
  - It is recorded systematically
  - Related to general propositions rather than being presented as reflecting a set of interesting curiosities
  - Is subjected to checks and controls on validity and reliability

## Direct vs. indirect observation

**Direct observation** occurs when the observer is physically present and personally monitors what takes place.

- ▶ This approach is very flexible because it allows the observer to react to and report subtle aspects of events and behaviors as they occur.
- ▶ The observer is free to shift places, change the focus of the observation or concentrate on unexpected events if they occur.

# **Indirect observation (mechanical observation)**

- ▶ Occurs when the recording is done by mechanical, photographic or electronic means.
- ▶ Indirect observation is less flexible than direct observation but is also less biasing and may be less erratic in accuracy.
- ▶ Another advantage of indirect observation is that the permanent record can be reanalyzed to include many different aspects of an event.

## **D. Focus groups**

- ▶ A focus group is a small discussion group without a fixed questionnaire.
- ▶ It is unstructured without a fixed questionnaire. It is unstructured so that spontaneous thoughts and ideas can surface.
- ▶ A focus group moderator is a person trained to conduct an unstructured discussion on a pre-specified list of topics.

## **E. Secondary Data Collection**

- A researcher gather secondary data from published and unpublished materials.
-

## **Analysis and Presentation of Data**

- Data analysis involves critical thinking.
- Here the research compiles or summarizes the results in a logical order usually in relation to the hypotheses and objectives of the study.
- After a brief introduction on the main results or features of the findings of the study, the data are described and interpreted in sufficient detail leading to the ultimate conclusion.
- Tables, graphs and illustrations are used to present the data more clearly and economically.

## Cont'd...

- One method of organizing the data analysis is to state each null hypothesis and to follow it with a description statistical test and interpretation of the results of the test.
- It involves depicting major results, evaluate and interpret them in light of the theoretical expectations implied in the hypothesis.
- The interpretation should bring out the meaning and implications of the results of the study and it should be supported with reference to related findings you recorded from literature review.

# Data analysis procedures

## 1. Editing

- Editing is the process of checking and adjusting data for omissions, consistency, and legibility. In this way, the data become ready for analysis by a computer.
- **Field Editing or in house-editing**

## 2. Coding

- Editing may be differentiated from **coding**, which is the assignment of numerical scores or classifying symbols to previously edited data. Careful editing makes the coding job easier. Codes are meant to represent the meaning in the data.

### **3. Classification:**

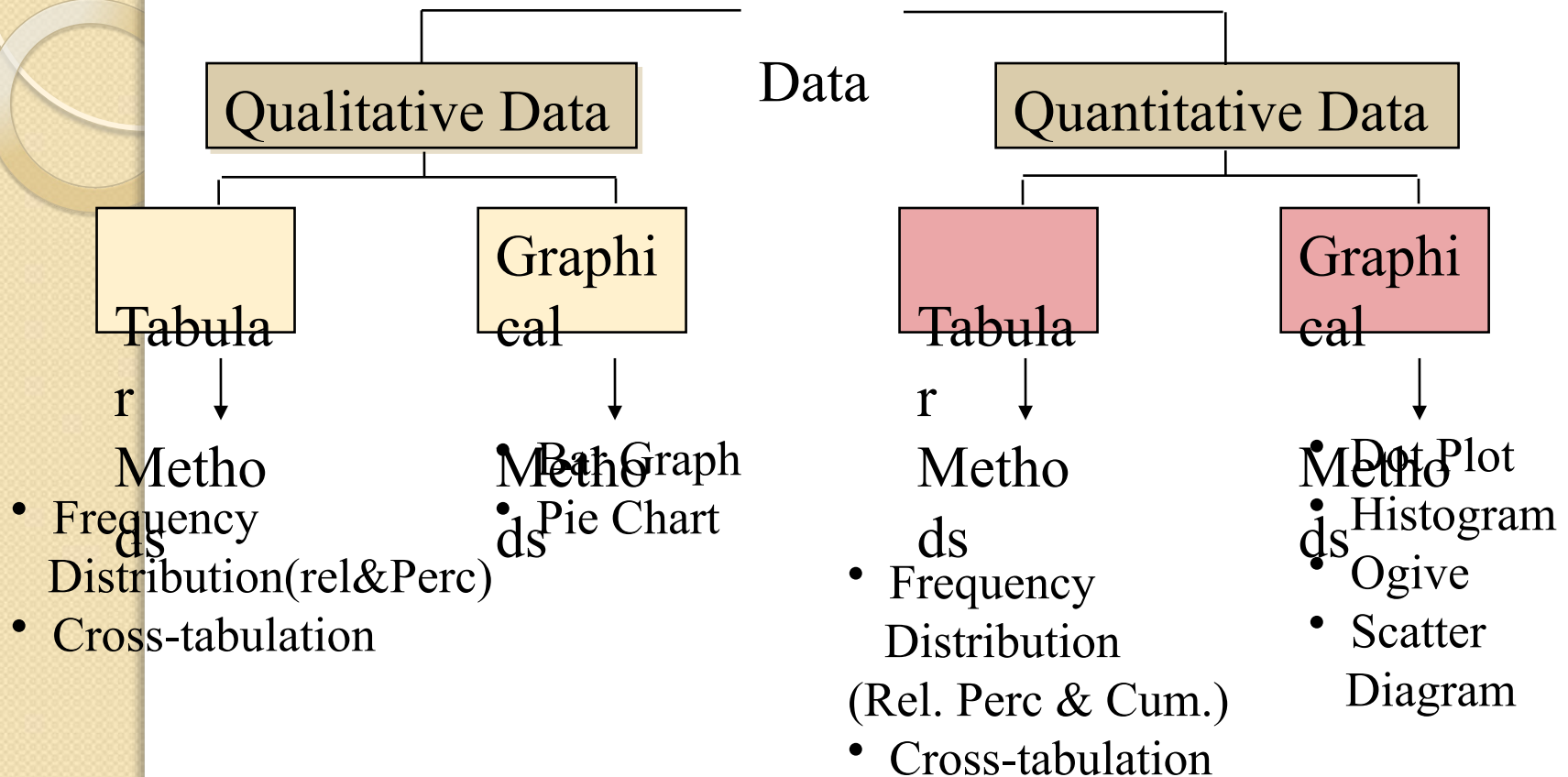
- Classification is the process of arranging data in groups or classes on the basis of common characteristics. Data having common characteristics are placed together and in this way the entire data gets divided into a number of groups or classes.
- Classification can be in the form of either attribute (literacy, sex, honesty etc) or class intervals (like income, population, age, weight etc).

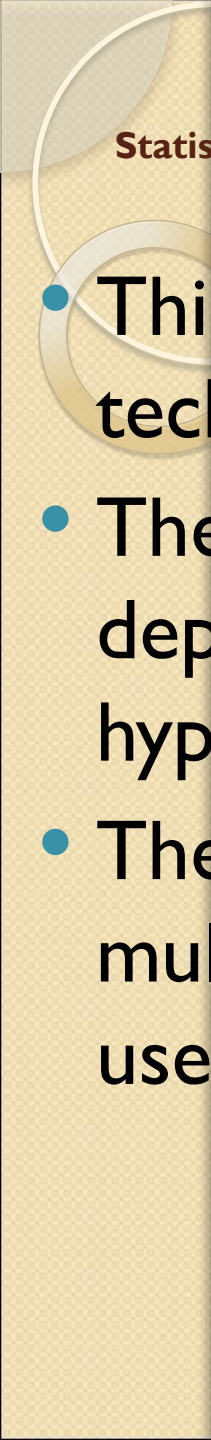
# Cont'd...

In the analysis of the data, you are expected to do the following:

- Classify data according to their similarity
- Describe the key variables as observed in the sample of the study.
- Estimate and discuss the population parameters from the sample statistics
- Describe the relationships between (or among) the variables for the sample. Estimate the degree of relationships between (or among) the variables in the population.
- Draw generalizations based upon the above analysis and the objectives of the study

# Summarizing & presenting data





### Statistical model:

- This requires choosing the appropriate statistical techniques to be used.
- The choice of the method of statistical analysis depends on the research problem and research hypothesis/objective:
- The number of variables (univariate, bivariate & multivariate) and scale of data measurement also used to determine statistical tools used.

- **Univariate data analysis:** When a researcher generalizes from a sample about one variable
- **Bivariate data analysis:** When the desire is to explain the relationship between two variables at a time
- **Multivariate data analysis:** Is the simultaneous investigation of more than two variables.
- **Scale measurement of data:** There are four scale measurements of data which were discussed in chapter seven (nominal, ordinal, interval and ratio scale).

## **6.4. Interpretation & conclusion/recommendation**

### **Interpretation:**

- Place under or next to data presentation by using tabular, graphical or other techniques.
- Data are interpreted, drawing implications of data & inferring based on nature and meaning of analyzed data.
- Care is needed not to undermine or overestimate the meaning of data.
- It needs validating your interpretation by previous studies (but no matter whether it support or contradict your findings....)
- Implications explained in non-jargon language, or lay terms.

## Conclusions and Recommendations

- In this chapter the researcher summarizes the main findings of his study and the implications.
- Conclusions summarize the main results of the research and describe what they mean for the general field.
- This section is not the place to mention new results for the first time.
- Avoid abbreviations, avoid over-extrapolation (of study finding), acronyms or citations.
- Briefly describe what you did, consider suggesting future research to follow up where your research ended.
- Provide readers with an interpretation of the impact of the research results, where appropriate.
- Provide finding based, professional, practical and specific recommendations.

## General format of Report Writing

- research report may differ considerably in scope of treatment and details of presentation.
- All types of research reports are expected to follow a common pattern /format.
- The general format of research report is evolved and it has become a tradition in academic area.
- It is viewed in three major categories:

# **Format of Research Report**

## **A. Preliminary Section**

1. Title page
2. Preface or acknowledgements
3. Table of content
4. List of tables (if any)
5. List of figures (if any).
6. Acronyms

# Cont'd...

## **B. Body of a Report**

### **Chapter one: Introduction**

- I.1. Background of the study
- I.2. Statement of the problem
- I.3. Objectives of the study
- I.4. Research questions/Hypotheses to be tested
- I.5. Significance of the study
- I.6. Scope of the study
- I.7. Limitation of the study
- I.8. Conceptual/operational definitions

## Chapter Two: Review of related literature

2.1. Theoretical review

2.2. Empirical review

## Chapter Three: Methodology

*3.1. Selection & description of study area*

*3.2. Research Strategy/design*

*3.3. Sources of data & methods of Collection*

*3.4. Sample size & sampling technique*

*3.5. Methods of Data processing and analysis.*



## Chapter Four: Analysis and presentation of data

- Tables and interpretation
- Figures and interpretation.

## Chapter Five: Conclusions & recommendation

- Main Findings, implication and inferences
- recommendations
- Suggestions for further studies.

## C. Reference

- It should be written in clear and consistent way where the researcher is expected to use each referencing system throughout the whole research.
- APA and Harvard referencing system are the widely used ones.

## **Appendix/Annex**

- The last is appendix where additional parts such as questionnaires, interview questions, relevant secondary data, pictures etc. are placed.

*End of the Course!!*